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TROCAR DEVICE AND METHOD

Abstract

A trocar assembly and associated methods are shown. Using example devices and methods shown, a surgeon may insert an obturator with a blade portion and a tube portion of a cannula into tissue such as eye tissue. In one state, the blade portions are wider than or equal to outer diameters of the tube portions. This facilitates an incision in tissue such as eye tissue that is wide enough to allow the tube portions to pass through. Because the blade portions are adapted to deform, they may be withdrawn through the inner diameter of the tube portions.

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Background/Summary

CLAIM OF PRIORITY [0001] This application is a continuation of and claims the benefit of priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 17/147,250, filed on Jan. 12, 2021, which is a continuation of and claims the benefit of priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 16/092,719, filed on Oct. 10, 2018, which is a U.S. National Stage Filing under 35 U.S.C. 371 from International Application No. PCT/US2017/027202 A1, filed on Apr. 12, 2017, and published as WO 2017/180739 A1 on Oct. 19, 2017, which claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 62/321,571, filed on Apr. 12, 2016, each of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD
[0002] This invention relates to devices and methods for insertion of a cannula.
BACKGROUND
[0003] A number of ophthalmological procedures require insertion of a cannula into an eye. Use of a cannula allows insertion and removal of instruments through a single opening, without making additional incisions. Existing technologies for insertion of a cannula use a blade to make an incision, however the blade is often smaller than an inner diameter of the cannula being placed. This can lead to a less precise fit between the incision and the cannula than is desired. Improved devices and procedures are desired to place cannulas into an eye.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0004] FIG. **1**A-**1**C shows a trocar device according to the prior art.
[0005] FIG. **2** shows a trocar device according to an embodiment of the invention.
[0006] FIG. **3**A-**3**B shows a trocar device blade according to an embodiment of the invention.
[0007] FIG. **4**A-**4**C shows a trocar device blade according to an embodiment of the invention.
[0008] FIG. **5**A-**5**C shows a trocar device blade according to an embodiment of the invention.
[0009] FIG. **6**A-**6**B shows a trocar device blade according to an embodiment of the invention.
[0010] FIG. **7** shows an example method of inserting a trocar according to an embodiment of the invention.
[0011] FIG. **8** shows another example method of inserting a trocar according to an embodiment of the invention.
DETAILED DESCRIPTION
[0012] In the following detailed description, reference is made to the accompanying drawings which form a part hereof, and in which is shown, by way of illustration, specific embodiments in which the invention may be practiced. In the drawings, like numerals describe substantially similar components throughout the several views. These embodiments are described in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments may be utilized and structural, or logical changes, etc. may be made without departing from the scope of the present invention.
[0013] FIG. **1**A shows an example of a cannula **100**. The cannula includes a tube **102** and a cap **104**. FIG. **1**B shows the cannula **100** when used with a trocar placement tool **110**. The obturator

may be used to place the cannula **100** into an eye or other cavity for use in a surgical procedure. The trocar placement tool **110** includes an obturator **112** and a handle **116**. In the example shown, the obturator **112** includes a tip **114**. In the example shown, the tip **114** includes a blade that is used to make an incision to place the cannula **100**. FIG. **1**C shows an example with the cannula **100** inserted into an eye **120**. The tube portion **102** of the cannula **100** is placed within the eye **120**, while the cap portion **104** remains outside the eye **120** and butts up against an outer surface of the eye **120**.

[0014] FIG. **2** shows another example of a trocar placement tool **200**. In the example of FIG. **2**, the trocar placement tool **200** includes a handle **202**, and a gripper **204** that may be used to hold a cannula **220**. Although a single gripper **204** is shown, two or more grippers may also be included. In the example shown, the cannula **220** is similar in size and shape to the example cannula **100** shown in FIGS. **1**A-**1**C. The cannula **220** is shown with a tube portion **222** and a cap portion **224**.

[0015] An obturator **210** is shown in FIG. **2** with a portion **214** that has an outer diameter sized to fit within an inner diameter of the cannula **220**. The obturator **210** of FIG. **2** further shows a blade **212** at a tip of the obturator **210**. The blade **212** is shown with a width that is wider than the inner diameter of the tube portion **222**. In the example shown, the blade **212** further includes a width that is wider than an outer diameter of the tube portion **222** of the cannula **220**. In one example, the blade **212** is a flexible blade as described in examples below.

[0016] FIG. **3**A shows one example of a trocar assembly **300** according to an embodiment of the invention. The trocar assembly **300** is shown in a first state that may be used for insertion of a cannula. The trocar assembly **300** includes a shaft portion **302** and a blade portion **304** of an obturator. The blade portion **304** is shown having a first width **306** in an non-deformed state. FIG. **3**A further shows a tube portion **310** of a cannula located around the shaft portion **302**. In the first state, as illustrated in FIG. **3**A, the first width **306** of the blade **304** is wider than an inner diameter **312** of the tube portion **310**. In one example in the first state, as illustrated in FIG. **3**A, the first width **306** of the blade **304** is also wider than an outer diameter **313** of the tube portion **310**. In one example (not shown) in the first state the first width **306** of the blade **304** may be equal to an outer diameter **313** of the tube portion **310**.

[0017] In the example shown, the blade **304** includes a cut out **308** that enhances an ability for the blade **304** to deform in a controlled manner. FIG. **3**B shows the trocar assembly **300** in a second state during removal of the obturator. The blade **304** is in a deformed state where the blade **304** now exhibits a second width **307**. As illustrated in FIG. **3**B, the second width is smaller or equal to the inner diameter **312** of the tube portion **310**. Although any of a number of geometries of cut outs **308** may be used, in the example shown in FIG. **3**B, the cut out **308** deforms at a location **309** to allow a wide portion of the blade **304** to deform to within the second width **307**.

[0018] In one example, the cut out **308** facilitates deformation of the blade portion **304** from the first width **306** to the second width **307**. Other examples may include a flexible blade portion **304** that does not utilize cut outs to enhance deformation, but otherwise utilizes thin blade material to permit deformation. In the example shown in FIG. **3**B, the deformation of the blade **304** is axial deformation. Other types of deformation may be used in other embodiments of the invention, as discussed below and shown in additional figures.

[0019] FIG. **4**A shows another example of a trocar assembly **400** according to an embodiment of the invention. The trocar assembly **400** is shown in a first state that may be used for insertion of a cannula. The trocar assembly **400** includes a shaft portion **402** and a blade portion **404** of an obturator. The blade portion **404** is shown having a first width **406** in an non-deformed state. FIG. **4**A further shows a tube portion **410** of a cannula located around the shaft portion **402**. In the first state, as illustrated in FIG. **4**A, the first width **406** of the blade **404** is wider than an inner diameter **412** of the tube portion **410**. In one example in the first state, as illustrated in FIG. **4**A, the first width **406** of the blade **304** is also wider than an outer diameter **413** of the tube portion **410**. In one example (not shown) in the first state the first width **406** of the blade **404** may be equal to an outer diameter **413** of the tube portion **410**.

[0020] In the example shown, the blade **404** includes a multiple cut outs **408** that enhance an ability for the blade **404** to deform in a controlled manner. FIG. **4**B shows the trocar assembly **400** in a second state during removal of the obturator. The blade **404** is in a deformed state where the blade **404** now exhibits a second width **407**. As illustrated in FIG. **4**B, the second width is smaller or equal to the inner diameter **412** of the tube portion **410**. In contrast to the example of FIGS. **3**A-**3**B, the cut outs **408** of blade **404** facilitate flexion upwards (or downwards) out of a plane of the blade **304**. FIG. **4**C shows an end view of the blade **404** with portions **405** is a deformed state, flexed out of a plane of blade **404**.

[0021] Similar to the example of FIGS. **3**A-**3**B, examples of the invention are not limited to configurations that include cut outs **408**. Other examples may include a flexible blade portion **404** that does not utilize cut outs to enhance deformation, but otherwise utilizes thin blade material to permit deformation.

[0022] FIG. **5**A shows one example of a trocar assembly **500** with a flexible blade that does not include a cutout according to an embodiment of the invention. The trocar assembly **500** is shown in a first state that may be used for insertion of a cannula. The trocar assembly **500** includes a shaft portion **502** and a blade portion **504** of an obturator. The blade portion **504** is shown having a first width **506** in an non-deformed state. End view illustration **508** also shows the non-deformed state.

[0023] FIG. **5**A further shows a tube portion **510** of a cannula located around the shaft portion **502**. In the first state, as illustrated in FIG. **5**A, the first width **506** of the blade **504** is wider than an inner diameter **512** of the tube portion **510**. In one example in the first state, as illustrated in FIG. **5**A, the first width **506** of the blade **504** is also wider than an outer diameter **513** of the tube portion **510**. In one example (not shown) in the first state the first width **506** of the blade **504** may be equal to an outer diameter **513** of the tube portion **510**.

[0024] End view illustration **509** shows the blade **504** in a second state during removal of the obturator. The blade **504** is in a deformed state where the blade **504** now exhibits a second width **507**. The second width **507** is smaller or equal to the inner diameter **512** of the tube portion **510**. The example of FIGS. **5**A-**5**C shows another example of flexion out of a plane of the blade **304**. End view **509** illustrates a distributed flexing gradient across substantially all of the blade **504**. The blade **504** is shown with a substantially smooth curve.

[0025] FIG. **5**B shows an end view of the blade **504** that further illustrates the tube portion **510**. In FIG. **5**B, the blade **504** is in an non-deformed state, and the end view **508** is wider than an inner diameter **512** of the tube portion **510**. In this example the blade **504** is also wider that the outer diameter **513** of the tube portion **510**. FIG. **5**C shows an end view of the blade **504** in a deformed state, flexed out of a plane of blade **504**, and the end view **509** is thinner than the inner diameter **512** of the tube portion **510**.

[0026] In operation, a surgeon may insert an obturator with a blade portion similar to blade portions **304**, **404**, and **504** and shaft portions similar to shaft portions **310**, **410**, and **510** into tissue such as eye tissue. In one state, as described above, the blade portions **304**, **404**, and **504** are wider than or equal to outer diameters of the tube portions **310**, **410**, and **510**. This facilitates an incision in tissue such as eye tissue that is wide enough to allow the tube portions **310**, **410**, and **510** to pass through. Because the blade portions **304**, **404**, and **504** are flexible, and are adapted to deform, they may be withdrawn through the inner diameter of the tube portions **310**, **410**, and **510**.

[0027] FIGS. **6**A and **6**B show a trocar assembly **600** according to another example of the invention. The assembly **600** includes a shaft portion **602** and a blade portion **604** of an obturator. In FIG. **6**A, the blade portion **604** is shown having a first width **607** in an non-deformed state. In FIGS. **6**A and **6**B, the non-deformed state provides the first width **607** that is smaller or equal to an inner diameter **612** of a tube portion **610**.

[0028] In the example of FIG. **6**A, the blade portion **604** includes an internal space **608** that is normally compressed at location **609**. The compression at location **609** allows the width **607** of the blade portion **604** to be smaller or equal to the inner diameter **612** of the tube portion **610**. Although one possible configuration of a compression **609** is shown, other locations and/or configurations of compression features or constrictions are within the scope of the invention.

[0029] As in other examples presented above, the blade portion **604** is a flexible blade portion. FIG. **6**A further shows a dilator **650** adapted to fit within the internal space **608** within the blade portion **604**. As a tip **652** of the dilator **650** enters the internal space **608** of the blade portion **604**, the internal space **608** is expanded to a deformed state shown in FIG. **6**B. In FIG. **6**B, with the dilator **650** inserted within the internal space **608**, the blade portion **604** is deformed or otherwise expanded to a second width **606**. As illustrated in FIG. **6**B, the second width **606** of the blade **604** is wider than an inner diameter **612** of the tube portion **610**. In one example, as illustrated in FIG. **6**B, the second width **606** of the blade **604** is also wider than an outer diameter **613** of the tube portion **610**. In one example (not shown) the second width **606** of the blade **604** may be equal to an outer diameter **613** of the tube portion **610**.

[0030] In operation, a surgeon may insert the obturator with the blade portion **604** and the shaft portion **620** inside the tube portion. Because the blade portion is in the non-deformed state shown in FIG. **6**A, the obturator will slide through the tube portion **610**. Then the dilator **650** may be inserted into the internal space **608** of the blade portion **604**. This will deform the blade portion **604** to the second width **606** as shown in FIG. **6**B. Now the blade portion **604** is wider than or equal to an outer diameter **613** of the tube portion **610**. This facilitates an incision in tissue such as eye tissue that is wide enough to allow the tube portion **610** to pass through.

[0031] The surgeon may then remove the dilator **650** from within the blade **604** of the obturator. The blade **604** will once again return to the width **607** that is smaller or equal to the inner diameter **612** of the tube portion **610**. The obturator and the blade **604** will thus be easily removed through the tube portion.

[0032] This procedure and the devices described allow a procedure to make a wide incision as a result of the temporary increased width **606**. This procedure and the devices described also allow the obturator to be withdrawn from within the tube portion **610** with a decreased width **607** after making the incision.

[0033] FIG. **7** shows an example method according to one example. In operation **702**, a trocar assembly is pushed into a tissue, the trocar assembly including a flexible blade having a width that is larger than an inner diameter of a cannula. In operation **704**, an incision is formed in the tissue wider than the inner diameter of the cannula using the flexible blade and guiding the cannula into the tissue through the incision. In operation **706**, the obturator and the blade are withdrawn, leaving the cannula in place. In operation **708**, the flexible blade is deformed while withdrawing the obturator and the blade such that the deformed blade has a width that is equal to or smaller than the inner diameter of the cannula.

[0034] FIG. **8** shows another method according to one example. In operation **802**, a trocar assembly is pushed into a tissue, the trocar assembly including a flexible blade at a tip of the obturator. In operation **804**, an incision is formed in the tissue wider than the inner diameter of the cannula using the flexible blade and guiding the cannula into the tissue through the incision. In operation **706**, a dilator is withdrawn, causing the flexible blade to have a second width that is equal to or less than the inner diameter of the cannula. In operation **708**, the obturator and the blade are withdrawn, leaving the cannula in place.

[0035] To better illustrate the method and apparatuses disclosed herein, a non-limiting list of embodiments is provided here:

[0036] Example 1 includes a trocar assembly. The assembly includes a cannula, an obturator within the cannula, and a flexible blade at a tip of the obturator, the flexible blade having a width that is larger than an inner diameter of the cannula, wherein the flexible blade is adapted to deform sufficiently to allow the obturator to be withdrawn through the inner diameter of the cannula.

[0037] Example 2 includes the trocar assembly of example 1, wherein the flexible blade is

substantially flat.
[0038] Example 3 includes the trocar assembly of any one of examples 1-2, wherein the width is larger than the outer diameter of the cannula.
[0039] Example 4 includes the trocar assembly of any one of examples 1-3, wherein the flexible blade is adapted to curl sufficiently to allow the obturator to be withdrawn through the inner diameter of the cannula.
[0040] Example 5 includes the trocar assembly of any one of examples 1-4, wherein the flexible blade is adapted to compress axially to allow the obturator to be withdrawn through the inner diameter of the cannula.
[0041] Example 6 includes the trocar assembly of any one of examples 1-5, wherein the flexible blade includes one or more cut outs within the blade to enhance flexing.
[0042] Example 7 includes the trocar assembly of any one of examples 1-6, wherein the flexible blade includes a central cut out within the blade to enhance flexing.
[0043] Example 8 includes the trocar assembly of any one of examples 1-7, wherein the flexible blade includes a pair of side cut outs within the blade to enhance flexing.
[0044] Example 9 includes the trocar assembly of any one of examples 1-4, wherein the flexible blade includes a steel blade.
[0045] Example 10 includes a trocar assembly. The assembly includes a cannula, an obturator within the cannula, a flexible blade at a tip of the obturator, the flexible blade having an internal space within the blade, and a dilator adapted to fit within the internal space within the blade, wherein a first width of the blade is equal to or wider than an inner diameter of the cannula when the dilator is inserted within the internal space, and wherein a second width of the blade is equal to or smaller than the inner diameter of the cannula when the dilator is removed from the internal space.
[0046] Example 11 includes the trocar assembly of example 10, wherein the obturator and the blade are integrally formed.
[0047] Example 12 includes the trocar assembly of any one of examples 10-11, wherein the blade is substantially flat.
[0048] Example 13 includes the trocar assembly of any one of examples 10-12, wherein the flexible blade includes a steel blade.
[0049] Example 14 includes a method of inserting a trocar. The method includes pushing a trocar assembly into a tissue, the trocar assembly including a cannula, an obturator within the cannula, and a flexible blade at a tip of the obturator, the flexible blade having a width that is larger than an inner diameter of the cannula. The method also includes forming an incision in the tissue wider than the inner diameter of the cannula using the flexible blade and guiding the cannula into the tissue through the incision, withdrawing the obturator and the blade, leaving the cannula in place, and deforming the flexible blade while withdrawing the obturator and the blade such that the deformed blade has a width that is equal to or smaller than the inner diameter of the cannula.
[0050] Example 15 includes the method of example 14, wherein deforming the flexible blade includes curling the flexible blade.
[0051] Example 16 includes the method of any one of examples 14-15, wherein deforming the flexible blade includes axially compressing at least a portion of the blade.
[0052] Example 17 includes a method of inserting a trocar. The method includes pushing a trocar assembly into a tissue, the trocar assembly including a cannula, an obturator within the cannula, a flexible blade at a tip of the obturator, and a dilator within the flexible blade causing the flexible blade to have a first width that is larger than an inner diameter of the cannula. The method also includes forming an incision in the tissue wider than the inner diameter of the cannula using the flexible blade and guiding the cannula into the tissue through the incision, withdrawing the dilator, causing the flexible blade to have a second width that is equal to or less than the inner diameter of the cannula, and withdrawing the obturator and the blade, leaving the cannula in place.

[0053] Example 18 includes the method of example 17, wherein withdrawing the dilator includes withdrawing the dilator from an internal space within obturator and the blade, wherein the internal space tapers to a more narrow space within the blade.
[0054] These and other examples and features of the present infusion devices, and related methods will be set forth in part in the above detailed description. This overview is intended to provide non-limiting examples of the present subject matter—it is not intended to provide an exclusive or exhaustive explanation.
[0055] The above detailed description includes references to the accompanying drawings, which form a part of the detailed description. The drawings show, by way of illustration, specific embodiments in which the invention can be practiced. These embodiments are also referred to herein as “examples.” Such examples can include elements in addition to those shown or described. However, the present inventors also contemplate examples in which only those elements shown or described are provided. Moreover, the present inventors also contemplate examples using any combination or permutation of those elements shown or described (or one or more aspects thereof), either with respect to a particular example (or one or more aspects thereof), or with respect to other examples (or one or more aspects thereof) shown or described herein.
[0056] In this document, the terms “a” or “an” are used, as is common in patent documents, to include one or more than one, independent of any other instances or usages of “at least one” or “one or more.” In this document, the term “or” is used to refer to a nonexclusive or, such that “A or B” includes “A but not B,” “B but not A,” and “A and B,” unless otherwise indicated. In this document, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Also, in the following claims, the terms “including” and “comprising” are open-ended, that is, a system, device, article, composition, formulation, or process that includes elements in addition to those listed after such a term in a claim are still deemed to fall within the scope of that claim. Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects.
[0057] The above description is intended to be illustrative, and not restrictive. For example, the above-described examples (or one or more aspects thereof) may be used in combination with each other. Other embodiments can be used, such as by one of ordinary skill in the art upon reviewing the above description. The Abstract is provided to comply with 37 C.F.R. § 1.72(b), to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Also, in the above Detailed Description, various features may be grouped together to streamline the disclosure. This should not be interpreted as intending that an unclaimed disclosed feature is essential to any claim. Rather, inventive subject matter may lie in less than all features of a particular disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment, and it is contemplated that such embodiments can be combined with each other in various combinations or permutations. The scope of the invention should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. A trocar assembly, comprising: a cannula; an obturator adapted to place the cannula; and a one piece flexible blade at a tip of the obturator, the flexible blade having a width that is larger than an inner diameter of the cannula, wherein the flexible blade includes one or more cutouts within the blade to enhance flexing.